

CSA0206 - C Programming

Assessment Tool - 01

JERLIN J

192312313

1. The company wants to automate its payroll process. Write a C program to calculate the total salary by adding the basic salary and bonus amount of an employee.

Input :

Basic Salary

Bonus Amount

Output:

Total Salary

Pseudocode :

Start

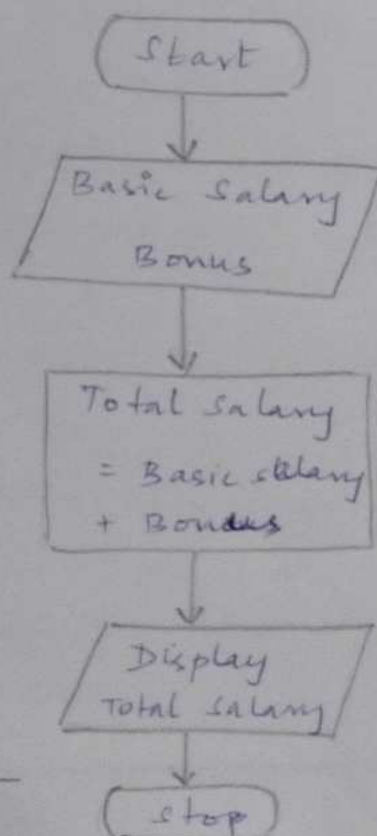
Read BasicSalary, Bonus

TotalSalary = BasicSalary + Bonus

Display TotalSalary

Stop

Flow chart :



C program:

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
float basic, bonus, total; scanf ("%f %f", &basic, &bonus)
```

```
printf ("Enter basic salary: ");
```

```
printf ("Enter bonus amount: ");
```

```
total = basic + bonus;
```

```
printf ("Total salary = %.2f", total);
```

```
return 0;
```

```
}
```

- 2). During a database update, two employee IDs need to be exchanged. Write a C program to swap two numbers using a temporary variable.

Input:

First Number (A)

Second Number (B)

Output:

Swapped values of A and B

Pseudocode:

Start

Read A, B

Temp = A

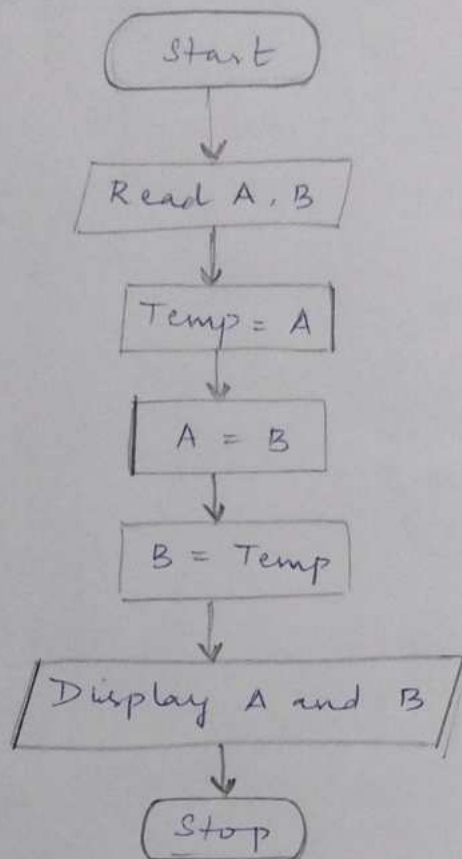
A = B

B = Temp

Display A, B

Stop

Flowchart:



C program:

```
#include <stdio.h>

int main ()
{
    int A, B, Temp;
    printf ("Enter two numbers: ");
    scanf ("%d %d", &A, &B);
    Temp = A;
    A = B;
    B = Temp;
    printf ("After Swapping: \n");
    printf ("A = %d\n", A);
    printf ("B = %d\n", B);
    return 0;
}
```

3). In an embedded system with limited memory, two values must be exchanged without using any extra storage. Write a C program to swap two numbers without using a temporary variable.

Input

First Number (A)

Second Number (B)

Output:

Swapped values of A and B

Pseudocode:

Start

Read A, B

$A = A + B$

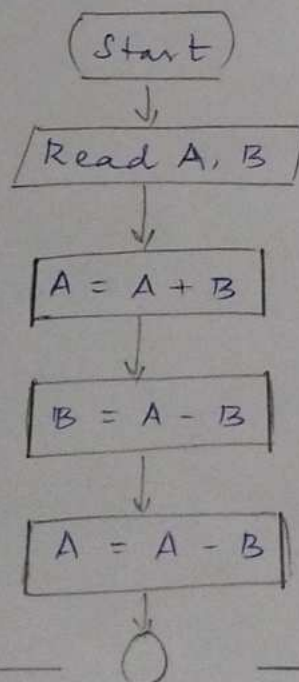
$B = A - B$

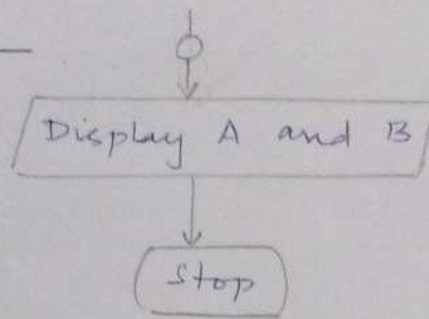
$A = A - B$

Display A, B

Stop

Flowchart :





C program:

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    int A, B ;
```

```
    scanf ("%d %d", &A, &B);
```

```
    printf ("Before swapping: A = %d B = %d", A, B);
```

```
    A = A + B;
```

```
    B = A - B;
```

```
    A = A - B;
```

```
    printf ("\n After swapping: A = %d B = %d", A, B);
```

```
    return 0;
```

```
}
```

4). A bank wants to provide customers with an estimate of the interest earned on their deposits. Write a C program to calculate simple interest using the formula.

Input:

Principal Amount (P)

Rate of Interest (R)

Time (T)

Output:

Simple Interest (SI):

Pseudocode:

Start

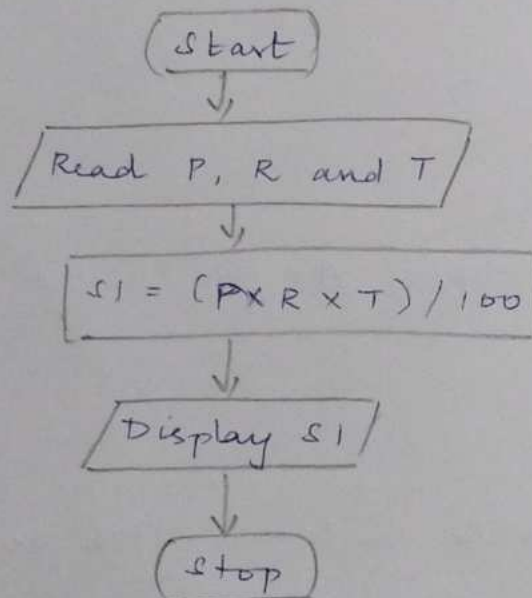
Read P, R, T

$SI = (P \times R \times T) / 100$

Display SI

stop

Flowchart:



C program..

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    float P, R, T, SI;
```

```
    scanf ("%f %f %f", &P, &R, &T);
```

```
    printf ("Enter Principal Amount: ");
```

```
    printf ("Enter Rate of Interest: ");
```

```
printf ("enter Time : ");
```

```
SI = (P * R * T) / 100;
```

```
printf ("Simple Interest = %f", SI);
```

```
return 0;
```

3.

- 5). A weather station records temperature in Celsius, but reports need to be generated in Fahrenheit. Write a program to convert temperature from Celsius to Fahrenheit.

Input:

Temperature in Celsius (C)

Output:

Temperature in Fahrenheit (F)

Pseudocode:

Start

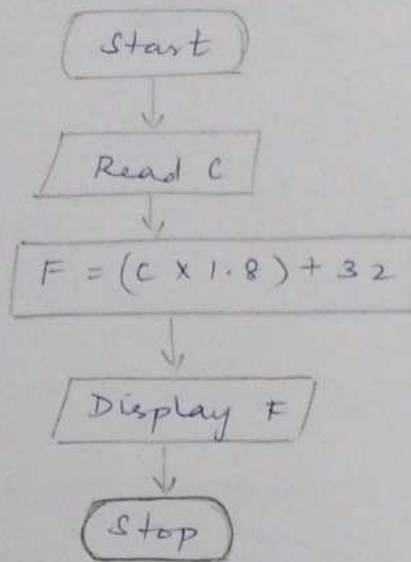
Read C

$F = (C \times 1.8) + 32$

Display F

Stop

Flow chart:



C program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float C, F;
```

```
    scanf("%f %f", &C, &F);
```

```
    F = (C * 1.8) + 32;
```

```
    printf("Temperature in celsius = %.2f", C);
```

```
    printf("Temperature in fahrenheit = %.2f", F);
```

```
    return 0;
```

```
}
```

- 6). A mathematics learning application needs to display both the quotient & remainder when two numbers are divided. Write a C program to find quotient & remainder of two integers.

Input:

First Integer (A)

Second Integer (B)

Output:

Quotient (Q)

Remainder (R)

Pseudocode:

start

Read A, B

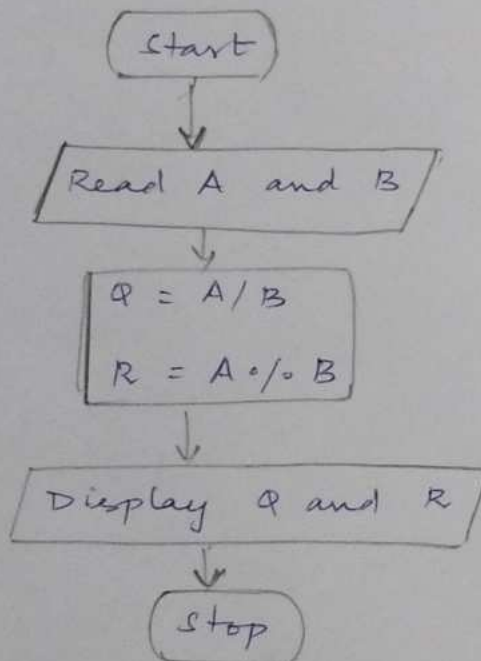
$Q = A / B$

$R = A \% B$

Display Q, R

stop

Flowchart:



C program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b;
```

```
    scanf ("%d %d", &a, &b);
```

```
printf("Quotient = %.d \n", a/b);  
printf("Remainder = %.d \n", a%b);  
return 0;  
}
```

7). An event management system allows entry based on even-numbered ticket IDs. Write a C program to determine whether a given ticket number is even or odd.

Input:

Ticket Number (N)

Output:

Even or Odd.

Pseudocode:

start

Read N

If $N \% 2 == 0$

Display "Even"

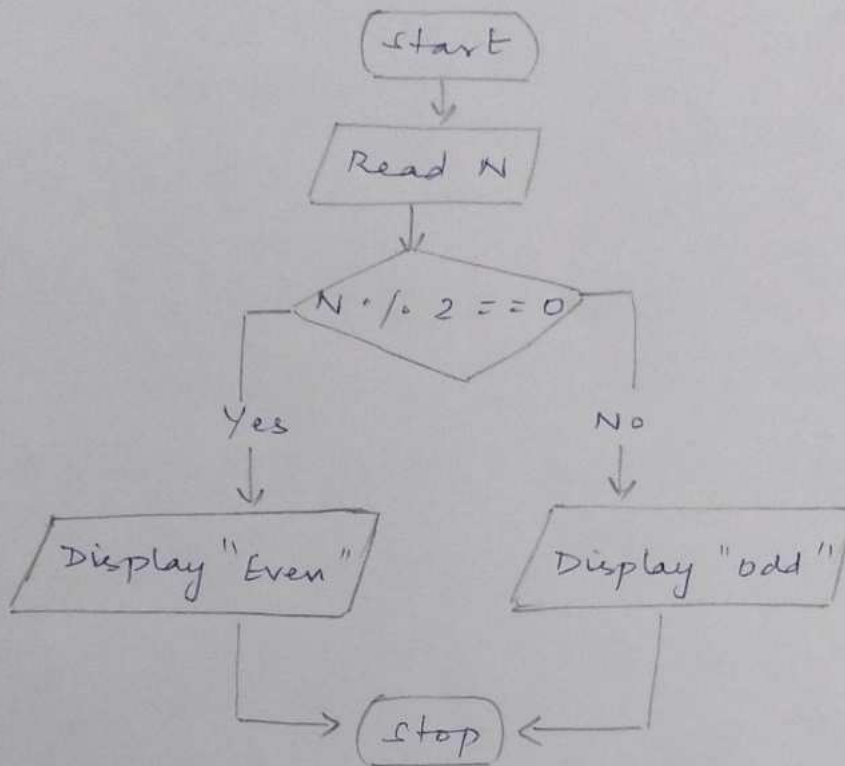
Else

Display "odd"

End If

stop

Flow chart:



C program:

```
#include <stdio.h>
int main()
{
    int num;
    scanf("%d", &num);
    if (num % 2 == 0)
        printf("%d is Even", num);
    else
        printf("%d is odd", num);
    return 0;
}
```

8). An engineer needs to solve quadratic equation of the form $ax^2 + bx + c = 0$. Write a C program to calculate and display the roots of quadratic equation.

Input:

Coefficient a

Coefficient b

Coefficient c

Output:

Root 1 (x_1)

Root 2 (x_2)

Pseudocode:

Start

Read a, b, c

$D = (b \times b) - (4 \times a \times c)$

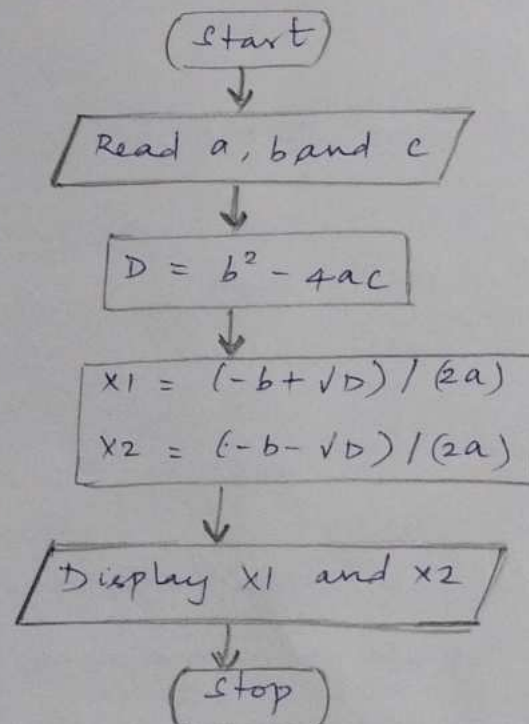
$x_1 = (-b + \sqrt{D}) / (2 \times a)$

$x_2 = (-b - \sqrt{D}) / (2 \times a)$

Display x_1 , x_2

Stop

Flowchart:



C program:

```
#include <stdio.h>

int main()
{
    float a, b, c, D, x1, x2;
    scanf ("%f %f %f", &a, &b, &c);
    printf ("Enter the values of a, b and c: ");
    D = (b*b) - (4*a*c);
    x1 = (-b + sqrt(D)) / (2*a);
    x2 = (-b - sqrt(D)) / (2*a);
    printf ("Root 1 = %.2f\n", x1);
    printf ("Root 2 = %.2f\n", x2);
    return 0;
}
```

9). A scientific calculator application requires the square and cube of a given number. Write a C program to calculate the square and cube of a number entered by the user.

Input:

Number (N)

Output:

Square of number

Cube of number

Pseudocode:

Start

Read N

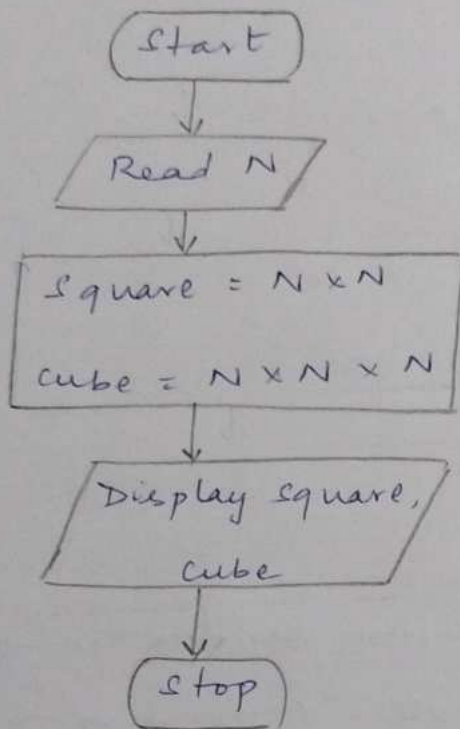
Square = $N \times N$

Cube = $N \times N \times N$

Display Square, cube

stop

Flowchart :



C program :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int N, square, cube;
```

```
    scanf("%d", &N);
```

```
    square = N * N;
```

```
    cube = N * N * N;
```

```
    printf("square = %d\n", square);
```

```
printf("cube = %.d\n", cube);
```

```
return 0;
```

```
}
```

- 10). A company wants to identify the top-performing salesperson among three employees based on their sales figures. Write a C program to find the largest number among three numbers.

Input:

First Number (A)

Second Number (B)

Third Number (C)

Output:

Largest Number.

Pseudocode :

Start

Read A, B, C

If $A > B$ and $A > C$ Then

Largest = A

Else If $B > C$

Largest = B

Else

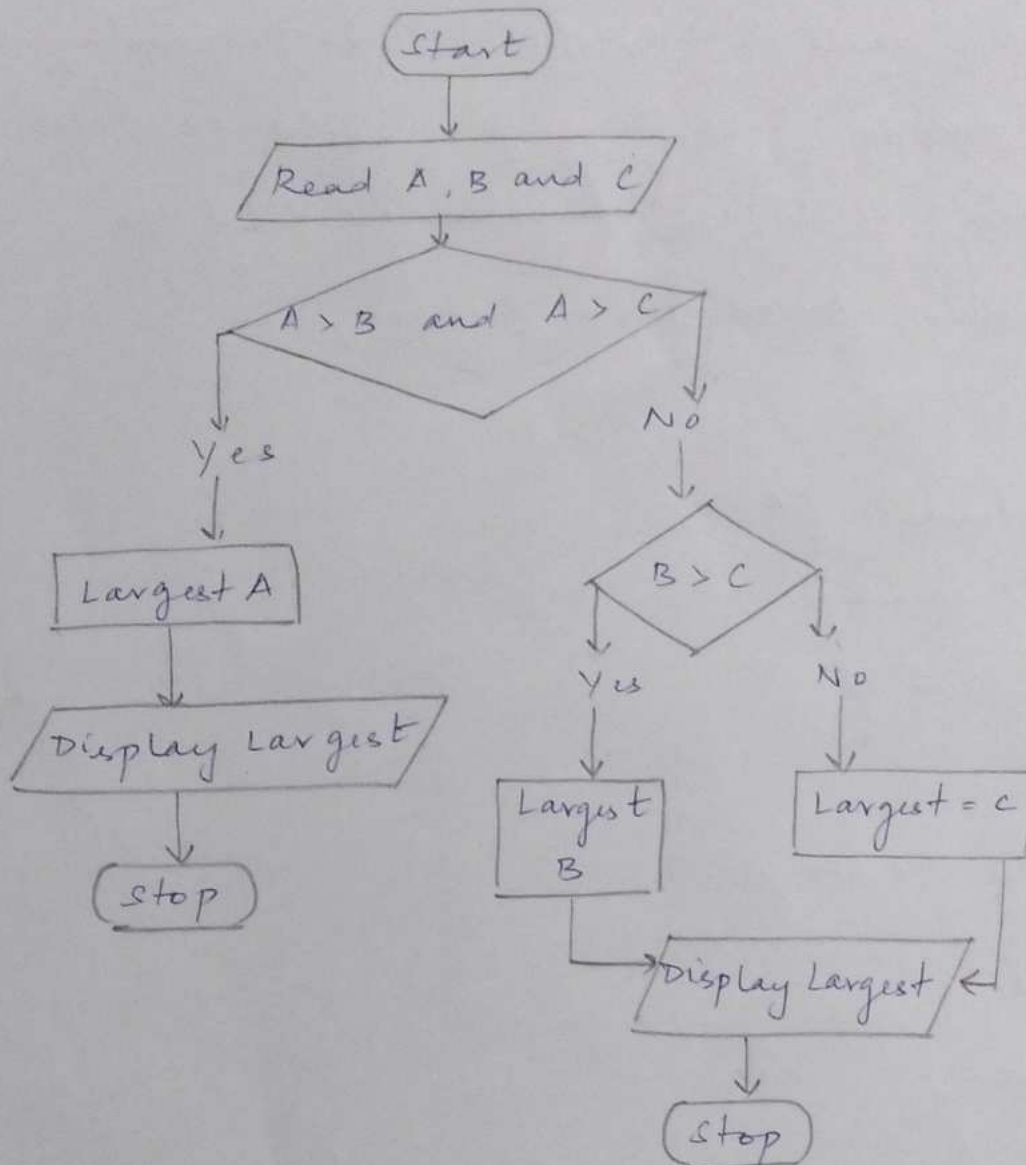
Largest = C

End If

Display Largest

stop

Flowchart:



C program:

```
#include <stdio.h>
int main()
{
    int A, B, C;
    scanf("%d %d %d", &A, &B, &C);
```

```
if (A > B && A > C)
```

```
{
```

```
    printf("Largest = %d", A);
```

```
}
```

```
else if (B > C)
```

```
{
```

```
    printf("Largest = %d", B);
```

```
else
```

```
{
```

```
    printf("Largest = %d", C);
```

```
}
```

```
return 0;
```

```
}
```